

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization. Software Developer and Data Engineer with 4+ years of professional experience and a completed Master of Science in Computer Science. Experienced in building scalable REST and SOAP microservices, cloud-native backend systems, and distributed data pipelines across healthcare, automotive, and research domains. Strong background in Java, Python, Spark, SQL, AWS, Azure, and CI/CD automation. Graduate work includes large-scale data platforms, infrastructure automation, and privacy-preserving LLM-based applications.

EXPERIENCE

NTT DATA

Software Developer

Bengaluru, India (Remote)

August 2022 – August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Built and automated CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and improving monitoring and error handling, reducing issue resolution time by 20 percent.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing.

Capgemini

Senior Analyst

Mumbai, India (Remote)

October 2021 – August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices.

Capgemini

Analyst

Mumbai, India (Remote)

July 2020 – October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage.

- Supported release planning and sprint execution using JIRA within Agile development teams.

PROJECTS

Central Michigan University

Graduate Student

Mount Pleasant, MI

August 2024 – December 2025

NYU Taxi Data Analysis Platform

Technologies: Apache Spark, PySpark, AWS, Docker, Terraform, Flask

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight.

Privacy-Preserving AI Email Client (Capstone Project)

Technologies: Python, PyTorch, Hugging Face, T5, LoRA

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under strict privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment.

EDUCATION

Central Michigan University

Master of Science in Computer Science

Mount Pleasant, MI

Aug 2024 – Dec 2025

GPA: 3.87 / 4.0

Relevant Coursework: Modern Databases, Pattern Recognition and Data Mining, Software and Data Modeling, Applied Data Engineering, Big Data Analytics, Cloud Computing, Analysis and Design of Algorithms.

Lovely Professional University

Bachelor of Technology in Computer Science and Engineering

Punjab, India

Aug 2016 – May 2020

CGPA: 8.02 / 10

TECHNICAL SKILLS

Java 8/11, Python, SQL, Spring Boot, Spring Framework, REST API, SOAP API, Apache Spark, PySpark, AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform, PostgreSQL, Oracle Database, IBM DB2, Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA,

Confluence, Tableau, DBeaver, ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals

Microsoft Certified: Azure Developer Associate

Microsoft Certified: Azure Data Engineer Associate